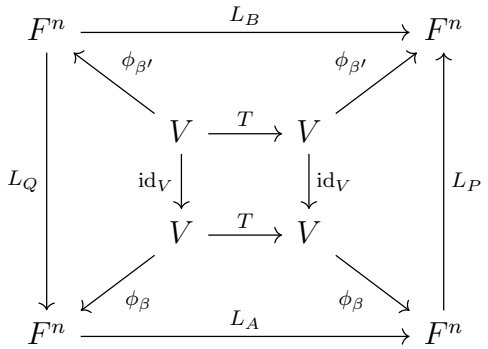
Linear Algebra I Recitation Class: Week 8

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the sets of rational, real, and complex numbers, respectively. V, W are vector spaces over a field F. If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind. If T is a linear transformation, we use the notation $\ker T := N(T)$ and $\operatorname{im} T := R(T)$.

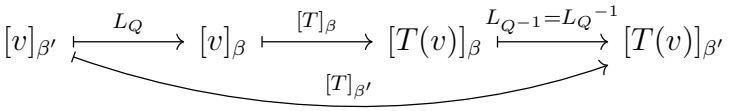
Let V be an n-dimensional vector space and $T : V \to V$ be a linear transformation. If β, β' are two ordered bases for V, we have the commutative diagram:



Here, $A = [T]_\beta, B = [T]_{\beta'}, Q = [\mathrm{id}_V]^\beta_{\beta'}, P = [\mathrm{id}_V]^{\beta'}_\beta = Q^{-1}$. Therefore,

$$B = PAQ = Q^{-1}AQ. \text{ That is, } [T]_{\beta'} = Q^{-1}[T]_\beta Q.$$

Q is called the **change of coordinate matrix** from β'-coordinates to β-coordinates. You can understand this equation by thinking that computing the matrix representation of a linear transformation in β'-coordinates involves first transforming the vector from β'-coordinates to β-coordinates, then applying the matrix representation with respect to β, and finally transforming the result back to β'-coordinates.



Let us do an example to get familiar with the computation:

Example 1 (Exercise 2.5.4)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$T\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2a + b \\ a - 3b \end{pmatrix}$$

Let β be the standard ordered basis for $\mathbb{R}^2$, and let

$$\beta' = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}.$$

Then

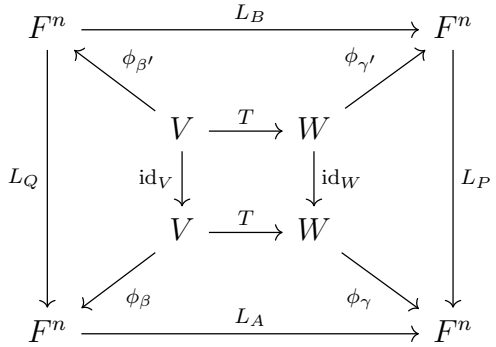
$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$
$$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} - 3\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Thus $[T]_\beta = \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix}$. Since $Q = [\mathrm{id}_V]^\beta_{\beta'} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, we have:

$$[T]_{\beta'} = [\mathrm{id}_V]^{\beta'}_\beta [T]_\beta [\mathrm{id}_V]^\beta_{\beta'} = Q^{-1} \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix} Q = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 13 \\ -5 & -9 \end{pmatrix}.$$

Change of coordinate matrices enable us to easily obtain a matrix representation of a linear transformation with various ordered bases. Some matrices are considered "nice" (for example, diagonal matrices). Naturally, we might ask if a linear transformation has a matrix representation that is a diagonal matrix. This gives us the motivation for diagonalization, which we will learn in the next semester.

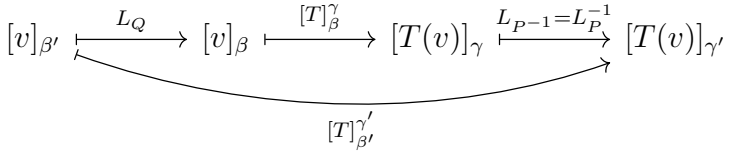
The notion of change of coordinates can be naturally generalized to linear transformations between two finite-dimensional vector spaces with different dimensions: if V and W are finite-dimensional vector spaces with dimensions n and m, and $T : V \to W$ is linear. Let β, β' be two ordered bases for V and γ, γ' be two ordered bases for W. Then we have the commutative diagram:



Here, $A = [T]^\gamma_\beta, B = [T]^{\gamma'}_{\beta'}, Q = [\mathrm{id}_V]^\beta_{\beta'}, P = [\mathrm{id}_W]^\gamma_{\gamma'}$. Therefore,

$$B = P^{-1}AQ. \text{ That is, } [T]^{\gamma'}_{\beta'} = P^{-1}[T]^\gamma_\beta Q.$$

In other words, any element $v \in V$ is sent by:



In the rest of the notes, we introduce a useful tool which will come in handy in the future (hopefully).

Definition 2. A **sequence** of vector spaces consists of a countable family of vector spaces V_i and linear transformations $f_i : V_i \to V_{i+1}$ for each $i \in \mathbb{Z}$. It is sometimes written as

$$\cdots \longrightarrow V_1 \xrightarrow{f_1} V_2 \xrightarrow{f_2} V_3 \xrightarrow{f_3} \cdots \xrightarrow{f_{n-1}} V_n \xrightarrow{f_n} V_{n+1} \longrightarrow \cdots.$$

Furthermore, the sequence is **exact at** V_n if $\ker f_n = \mathrm{im}\, f_{n-1}$. If the sequence is exact at every V_i, then we say the sequence is an **exact sequence**.

We are interested in sequences of the following form:

Definition 3. A **short exact sequence** of vector spaces is an exact sequence of the form

$$0 \longrightarrow U \xrightarrow{f} V \xrightarrow{g} W \longrightarrow 0$$

Here, 0 denotes the zero vector space. Since the map $0 \to U$ has image $\{0\}$, by the exactness at U, $\ker f = \{0\}$, that is, f is one-to-one. On the other hand, the map $W \to 0$ has kernel W, so by the exactness at W, $\operatorname{im} g = W$, that is, g is onto.

Example 4. Let W be a vector subspace of V. We have the linear map $\iota : W \to V$ defined by $\iota(w) = w$ for any $w \in W$; this is called the **inclusion** of W into V. On the other hand, recall that we have a canonical projection $\pi : V \to V/W$ defined by $\pi(v) := v + W$. Then we have an exact sequence:

$$0 \longrightarrow W \xrightarrow{\iota} V \xrightarrow{\pi} V/W \longrightarrow 0.$$

More generally,

Example 5. Let $T : V \to W$ be linear, it gives rise to the short exact sequence

$$0 \longrightarrow \ker T \xrightarrow{\iota} V \xrightarrow{T} \operatorname{im} T \longrightarrow 0.$$

Two commuting exact sequences have a nice property regarding their connecting maps:

Lemma 6 (The Short Five Lemma)**.** *Consider the following diagram of vector spaces*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & U & \xrightarrow{f} & V & \xrightarrow{g} & W & \longrightarrow & 0 \\ & & \alpha\downarrow & & \beta\downarrow & & \gamma\downarrow & & \\ 0 & \longrightarrow & A & \xrightarrow{f'} & B & \xrightarrow{g'} & C & \longrightarrow & 0 \end{array}$$

where the upper horizontal and the lower horizontal sequences are exact. Then:

1. *If α, γ are one-to-one, then β is one-to-one.*
2. *If α, γ are onto, then β is onto.*
3. *If α, γ are isomorphisms, then β is an isomorphism.*

Proof. 1. Let $v \in \ker \beta$; it suffices to prove that $v = 0$. Since $\beta(v) = 0$, by commutativity, $g'\beta(v) = \gamma g(v) = 0$. Thus $g(v) = 0$ by the injectivity of γ, i.e., $v \in \ker g = \operatorname{im} f$ by exactness. Say $v = f(u)$ for some $u \in U$. Then by commutativity again,

$$f'\alpha(u) = \beta f(u) = \beta(v) = 0.$$

By the injectivity of f' and α, we know that $u = 0$, thus $v = f(u) = 0$.

2. For any $b \in B$, we prove that there exists $x \in V$ such that $\beta(x) = b$. Since γ is onto, there exists $w \in W$ such that $\gamma(w) = g'(b)$, and for this w, there exists $v \in V$ such that $g(v) = w$ by the exactness of the sequence. By commutativity:

$$g'(b) = \gamma g(v) = g'\beta(v) \implies b - \beta(v) \in \ker g' = \operatorname{im} f'.$$

Say $f'(a) = b - \beta(v)$ for some $a \in A$, and $\alpha(u) = a$ for some $u \in U$ by the surjectivity of α. Define $x = f(u) + v$. Then $x \in V$, and

$$\beta(x) = \beta f(u) + \beta(v) = f'\alpha(u) + \beta(v) = b - \beta(v) + \beta(v) = b.$$

3. This is immediate from 1 and 2.

□

Remark. The technique of this proof is called "diagram chasing," namely, chasing an element through the diagram.

Exercise A (The Five Lemma)**.** Consider the following diagram of vector spaces:

$$\begin{array}{ccccccccc} T & \xrightarrow{e} & U & \xrightarrow{f} & V & \xrightarrow{g} & W & \xrightarrow{h} & X \\ \downarrow{\scriptstyle \eta} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} \\ Z & \xrightarrow{e'} & A & \xrightarrow{f'} & B & \xrightarrow{g'} & C & \xrightarrow{h'} & D \end{array}$$

where the upper horizontal and the lower horizontal sequences are exact. Show the following:

1. If η is onto and α, γ are one-to-one, then β is one-to-one.
2. If δ is one-to-one and α, γ are onto, then β is onto.
3. If $\eta, \alpha, \gamma, \delta$ are all isomorphisms, then β is an isomorphism.

Recall that for any linear transformation $T : V \to W$, we define the kernel as $\ker T := T^{-1}(\{0_W\})$. We now define the **cokernel** of T:

$$\operatorname{coker} T := W / \operatorname{im} T.$$

The reason the cokernel is called the "co"-kernel is that it is a "dual" notion to the kernel in a categorical sense, which we will not discuss just yet.

An immediate consequence is that T is onto if and only if $\operatorname{coker} T = \{0\}$. Are you getting déjà vu?

With the cokernel, for any linear transformation $T : V \to W$, we always have an exact sequence:

$$0 \longrightarrow \ker T \xrightarrow{\iota} V \xrightarrow{T} W \xrightarrow{\pi} \operatorname{coker} T \longrightarrow 0.$$

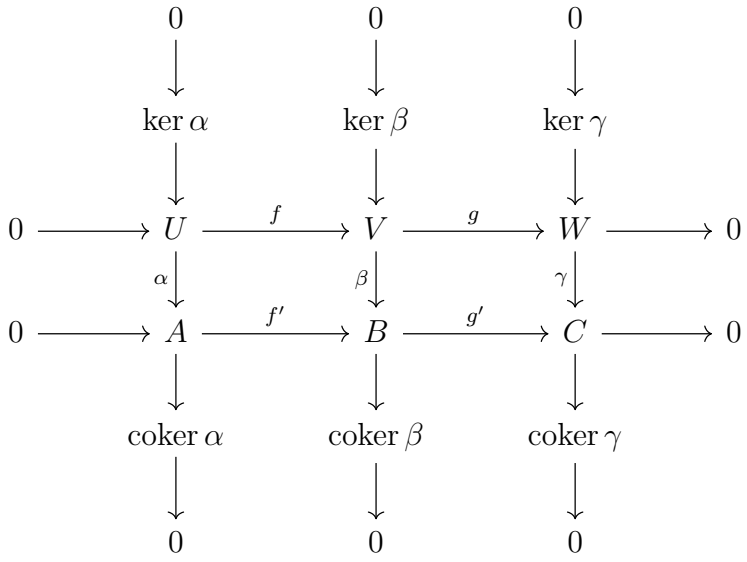
Here, ι is the inclusion, and π is the canonical projection.

The following lemma is an important and, moreover, famous result regarding exact sequences:

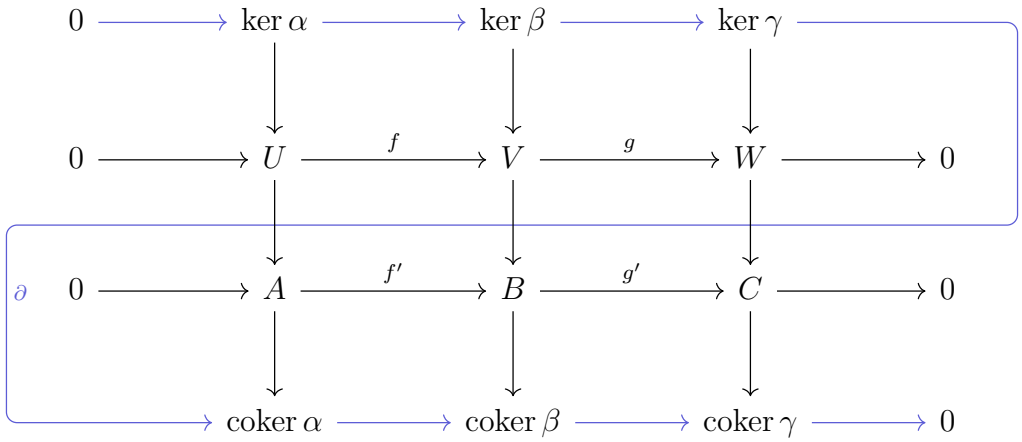
Lemma 7 (The Snake Lemma)**.** *Consider the following diagram of vector spaces:*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & U & \xrightarrow{f} & V & \xrightarrow{g} & W & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & A & \xrightarrow{f'} & B & \xrightarrow{g'} & C & \longrightarrow & 0 \end{array}$$

where the upper horizontal and the lower horizontal sequences are exact. Note that each vertical map induces the diagram:



Then we have an exact sequence in blue:



Remark. The map $\partial : \ker \gamma \to \operatorname{coker} \alpha$ is sometimes called the **connecting homomorphism**.

Proof. You can refer to [Bou98, I.1.4 Proposition 2] or try doing it yourself! It is a good exercise for diagram chasing. □

References

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[Rom08] Steven Roman. *Advanced linear algebra*, volume 135 of *Graduate Texts in Mathematics*. Springer, New York, third edition, 2008.